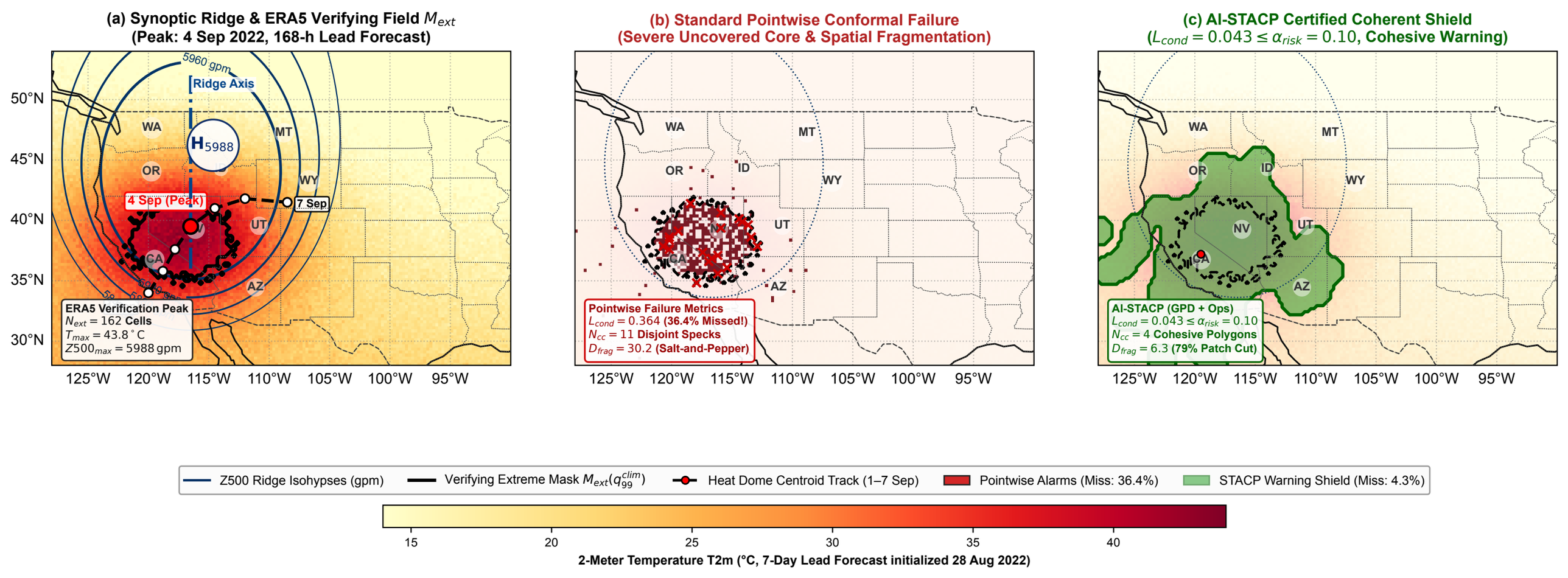

(a) Synoptic Ridge & ERA5 Verifying Field M_{ext}
(Peak: 4 Sep 2022, 168-h Lead Forecast)
5960 gpm
Ridge Axis
H5988
4 Sep (Peak)
7 Sep
ERA5 Verification Peak
N_{ext} = 162 Cells
T_{max} = 43.8 °C
$Z500_{max}$ = 5988 gpm
(b) Standard Pointwise Conformal Failure
(Severe Uncovered Core & Spatial Fragmentation)
Pointwise Failure Metrics
L_{cond} = 0.364 (36.4% Missed!)
N_{cc} = 11 Disjoint Specks
D_{frag} = 30.2 (Salt-and-Pepper)
(c) AI-STACP Certified Coherent Shield
(L_{cond} = 0.043 ≤ α_{risk} = 0.10, Cohesive Warning)
AI-STACP (GPD + Ops)
L_{cond} = 0.043 ≤ α_{risk} = 0.10
N_{cc} = 4 Cohesive Polygons
D_{frag} = 6.3 (79% Patch Cut)
WA
OR
CA
NV
ID
MT
WY
UT
AZ
50°N
45°N
40°N
35°N
30°N
125°W
120°W
115°W
110°W
105°W
100°W
95°W
Z500 Ridge Isohypses (gpm)
Verifying Extreme Mask $M_{ext}(q_{99}^{clim})$
Heat Dome Centroid Track (1–7 Sep)
Pointwise Alarms (Miss: 36.4%)
STACP Warning Shield (Miss: 4.3%)
15
20
25
30
35
40
2-Meter Temperature T2m (°C, 7-Day Lead Forecast initialized 28 Aug 2022)